Nishil Mehta

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EDUCATION

2023 – present
PhD Student - Study of the atmosphere of hot Neptunes
Université Côte d'Azur, Observatoire de la Côte d'Azur, CNRS, Laboratoire Lagrange, France
Supervisor: Vivien Parmentier, Tristan Guillot

2018 – 2023
BS-MS Dual Degree
Indian Institute of Science Education and Research, Thiruvananthapuram (IISER-TVM)
Major: Physics
Minor: Data Science

PUBLICATIONS

May 2024
Multiple Clues for Dayside Aerosols and Temperature Gradients in WASP-69 b from a Panchromatic JWST Emission Spectrum
arXiv:2406.15543
Everett Schlawin, Sagnick Mukherjee, Kazumasa Ohno, Taylor Bell, Thomas Beatty, Thomas Greene, Michael Line, Ryan Challener, Vivien Parmentier, Jonathan Fortney, Emily Rauscher, Lindsey Wiser, Luis Welbanks, Matthew Murphy, Isaac Edelman, Natasha Batalha, Sarah Moran, Nishil Mehta, and Marcia J. Rieke

Feb 2024
Exploring Gamma-Ray Burst Diversity: Clustering analysis of emission characteristics of Fermi and BATSE detected GRBs
The Astrophysical Journal (ApJ 969, 88)
Nishil Mehta, Shabnam Iyyani

CONFERENCE

Aug 2023
Strange New Worlds: The Exploration of Exoplanets, IISER Pune
Contributed Talk: The feasibility of detecting biosignatures in Archean Earth-like Trappist-1e exoplanet using JWST

RESEARCH EXPERIENCE

Nov 2023 – present
3D Modeling of the exoplanetary atmosphere using General Circulation Model - SPARC/MITgcm
Observatoire de la Côte d'Azur, Nice, France
Graduate Advisor: Vivien Parmentier

May 2023 – Nov 2023
Developing a 1D photochemistry/diffusion code for Exoplanetary Atmosphere
National Institute of Science Education and Research (NISER), Bhubaneswar
Guide: Liton Majumdar

A photochemical code with diffusion is being developed from scratch using Python for the atmospheres of exoplanets.

Aug 2022 – May 2023
Masters Thesis (Physics): The study of the Exoplanetary Atmosphere using the JWST and investigation of biosignatures. (Mehta et. al. in prep.)
Indian Institute of Science Education and Research, Thiruvananthapuram (IISER-TVM),
National Institute of Science Education and Research (NISER), Bhubaneswar
Undergraduate Advisor: Liton Majumdar (NISER), Soumen Basak (IISER-TVM)

Studying the atmosphere of exoplanets using forward modelling and retrieval methods.
A detailed investigation of biosignatures was conducted by developing an astrobiology model for archean Earth-like planets.

Jan 2022 – Jul 2022
Minor Thesis Project (Data Science): Exploring Gamma-Ray Burst Diversity: Clustering analysis of emission characteristics of Fermi and BATSE detected GRBs (ApJ 969, 88)
IISER-TVM
Undergraduate Advisor: Shabnam Iyyani

Created a Machine Learning classification model for Gamma-ray Bursts with minimal parameters that work for multiple catalogs (Fermi and BATSE).

Analyzed Gamma-ray Bursts data from telescopes such as Fermi, and BATSE.

Dec 2020 – Jul 2021 **Modelling of Astrophysical objects**
IISER-TVM
Guide: Liton Majumdar

Physical, chemical, and radiative transfer modelling of prestellar core L1544.

Jan 2020 – Dec 2020 **Astronomical Spectroscopy**
NISER
Guide: Liton Majumdar

Computation of the spectroscopic constants of molecules found in the Interstellar Medium using Quantum chemistry software and further analysis.

May 2019 – Jul 2019 **A Deep Learning Study of the Potential Energy Surfaces of Small Triangular Molecules.**
Summer Internship, IIT-Bombay
Guide: Alok Shukla

Quantum Calculations using Machine Learning (Neural Networks).
Reduced the computational cost by implementing machine learning (Neural Networks) to calculate the potential energy using fewer data and more accuracy.

PROFESSIONAL EXPERIENCE

Oct 2021 – Mar 2023 **Managed Network Expert**
Chegg India
Subject Expert: Advanced Physics (Freelance)

SKILLS

Programming
Python, C++, C, R, Fortran, Solidity (Basic)
Assembly Language (Intel 8085),
Mathematica and MATLAB

Technical Skills
Machine Learning, Neural Networks,
Forward Modeling and Retrieval Techniques (Exoplanetary Atmosphere)
Modeling: Astrobiology Model

Software
Ab initio software: Psi4, Quantum espresso, Gamess, Gaussian;
Gas-grain chemistry code: Nautilus
Radiative Transfer: gCMCRT, petitRADTRANS, Taurex
GCM: MITgcm and postprocessing

Data Analysis
SQL, ALMA archive data analysis using CASA

AWARD

INSPIRE (Innovation in Scientific Pursuit for Inspired Research) Scholarship
Department of Science and Technology (DST), Government of India.

LEADERSHIP EXPERIENCE

Co-Founder and Club President (2020-2021) of Parsec, The Astronomy Club of IISER-TVM.
Administered the work of the club and assisted in the development of the club website.
Organized events and handled social media for the club.
Handled and managed the telescope and star-gazing sessions.

Physics Expo Head
Oversaw the working of the Physics expo and assisted in organizing Anvesha, the annual science festival of IISER-TVM.

INTERESTS

Skating
- Represented India at *World Slalom Skating Association* at China.
- Represented India at Thailand Slalom Association and **secured 3rd rank**.

Programming
Organized Hackathons at IISER-TVM. (Code Battle: 2020–22, Error 404: 2019)

Cryptocurrency Enthusiast
Blockchain, DeFi, Metaverse, and Web3.0 with basic skills in Smart Contract Development.